MATHEMATICS BY D.K. SINGH
JEE MAINS MOCK TEST

Time: 1 Hour 15 minutes **Maximum Marks: 100**

Read the instructions carefully before answering:

1. This paper contains two sections: **Section 1** and **Section 2**.
2. **Section 1** consists of 20 Single Correct Choice Questions (Questions 1 to 20).
3. **Section 2** consists of 5 Integral/Numerical Answer Type Questions (Questions 21 to 25).
4. Filling up the responses carefully on the OMR sheet is mandatory.

Section 1: Single Correct Choice Type (Questions 1 to 20)

Marking Scheme:

- **Full Marks:** +4 if only the correct option is chosen.
- **Zero Marks:** 0 if none of the options are chosen.
- **Negative Marks:** -1 in all other cases.

1. If $A+B+C=\pi$ and $8\sin\left(A+\frac{C}{2}\right)=k\sin\frac{C}{2}$, then $\tan\frac{A}{2}\tan\frac{B}{2}=$

(A) $\frac{k-1}{k+1}$ (C) $\frac{k+1}{k-1}$
(B) $\frac{k+1}{k-1}$ (D) $\frac{k+1}{k}$

2. The value of $\sin 55^\circ - \sin 19^\circ + \sin 53^\circ - \sin 17^\circ$ is always equal to

(A) $\cot 1^\circ$ (C) $\tan 1^\circ$
(B) $\sin 1^\circ$ (D) $\cos 1^\circ$

3. Exact value of $\cos^2 73^\circ + \cos^2 47^\circ - \sin^2 43^\circ + \sin^2 107^\circ$ is equal to:

(A) 1/2 (C) 1
(B) 3/4 (D) None

4. The expression $\frac{1+\sin 2\alpha}{\cos(2\alpha-2\pi)\tan\left(\alpha-\frac{3\pi}{4}\right)} - \frac{1}{4}\sin 2\alpha\left[\cot\frac{\alpha}{2}+\cot\left(\frac{3\pi}{2}-\frac{\alpha}{2}\right)\right]$ is equal to

(A) 1 (C) $2\sin^2(\alpha/2)$
(B) 0 (D) $\sin^2\alpha$

5. The value of x satisfying the equation $2^{\log_3 x}+8=3\cdot x^{\log_9 4}$ is:

(A) an irrational number
(B) an odd prime number
(C) relatively prime with 4
(D) a rational number which is not an integer

6. If $\log_y x + \log_x y = 7$, then the value of $(\log_y x)^3 + (\log_x y)^3$ is

(A) 322
(B) 342
(C) 343
(D) 344

7. Let a, b, c be real numbers greater than 1. If $s = \log_a bc + \log_b ca + \log_c ab$, then

(A) $s \leq 9$
(B) $s \geq 9$
(C) $s \leq 6$
(D) $s \geq 6$

8. For a positive number m, characteristic of $\log_{10} m$−Mantissa of $\log_{10} m = 1.5$. Then $\log_{100}\left(\frac{m}{\sqrt{10}}\right) =$

(A) 1
(B) -1
(C) 0
(D) Insufficient data

9. The number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81$, $0 \leq x \leq \frac{\pi}{4}$ is:

(A) 1
(B) 3
(C) 0
(D) 2

10. If $x^2 + 9y^2 - 4x + 3 = 0$, where $x, y \in \mathbb{R}$, then x and y respectively lie in the intervals:

(A) $[1, 3]$ and $[1, 3]$
(B) $[1, 3]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$
(C) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $[1, 3]$
(D) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

11. The value of $2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is:

(A) $\frac{1}{4}$
(B) $\frac{1}{8\sqrt{2}}$
(C) $\frac{1}{8}$
(D) $\frac{1}{4\sqrt{2}}$

12. If n is the number of solutions of the equation $2\cos x\left(4\sin\left(\frac{\pi}{4} + x\right)\sin\left(\frac{\pi}{4} - x\right) - 1\right) = 1$, $x \in [0, \pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is:

(A) $\left(3, \frac{5\pi}{3}\right)$
(B) $(3, 2)$
(C) $\left(3, \frac{13\pi}{9}\right)$
(D) $\left(2, \frac{8\pi}{9}\right)$

13. The sum of solutions of the equation $\frac{\cos x}{1+\sin x} = |\tan 2x|$ is, where $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$:

(A) $\frac{\pi}{10}$
(B) $-\frac{11\pi}{30}$
(C) $\frac{\pi}{15}$
(D) $-\frac{7\pi}{30}$

14. A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with the lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is:

(A) $12\sqrt{15}$
(B) $12\sqrt{10}$
(C) $8\sqrt{10}$
(D) $6\sqrt{10}$

15. The sum of the roots of the equation, $x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is:

(A) $\log_2 12$
(B) $\log_2 13$
(C) $\log_2 11$
(D) $\log_2 14$

16. Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10}-2a_8}{3a_9}$ is:

(A) 2
(B) 3
(C) 1
(D) 4

17. The number of solutions of the equation, $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is:

(A) 5
(B) 2
(C) 4
(D) 3

18. The number of real values of x satisfying the equation $\log_{(x+1)}(2x^2+7x+5)+\log_{(2x+5)}(x+1)^2 = 4$ is:

(A) 0
(B) 1
(C) 2
(D) 3

19. The value of $(-\csc 6^\circ + \csc 18^\circ + \sec 24^\circ - \sec 36^\circ + \csc 42^\circ - \csc 78^\circ)$ is equal to:

(A) -6
(B) 6
(C) 4
(D) 2

20. If $0 < \theta < \frac{\pi}{2}$ and $f(\theta) = (1 + \cot\theta - \csc\theta)(1 + \tan\theta + \sec\theta)$, then $f\left(\frac{\pi}{12}\right)$ is equal to:

(A) 0
(B) 1
(C) $\sqrt{2}$
(D) 2

Section 2: Integral/Numerical Answer Type (Questions 21 to 25)

Marking Scheme:

- **Full Marks:** +4 if only the correct integer/numerical value is written.
- **Zero Marks:** 0 if the question is unanswered.
- **Negative Marks:** −1 in all other cases.

21. It is known that $\left|2\cos^3\frac{\pi}{7}-\cos^2\frac{\pi}{7}-\cos\frac{\pi}{7}\right|$ is a rational number. Write this rational number in the form p/q, where p and q are coprime natural numbers, then find the value of $p+q$.

22. If $\alpha,\beta,\gamma,\delta$ are the smallest positive angles in ascending order which have their sines equal to $\frac{7}{9}$, then find the value of $\left(12\sin\frac{\alpha}{2}+9\sin\frac{\beta}{2}+6\sin\frac{\gamma}{2}+3\sin\frac{\delta}{2}\right)$.

23. The number of integral values of 'a' such that both roots of the equation $ax^2+(3a+1)x-5=0$ are integers is:

24. $ABCDEFG$ is a 7-sided regular polygon whose side length is 1, then find the value of $\frac{1}{AC}+\frac{1}{AD}$.

25. If α,β are the roots of the equation $\lambda(x^2-x)+x+5=0$ and if λ_1,λ_2 are two values of λ that satisfy $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{4}{5}$, then find the value of $\frac{\lambda_1}{\lambda_2^2}+\frac{\lambda_2}{\lambda_1^2}$.

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OMR RESPONSE SHEET

CANDIDATE DETAILS:

Candidate's Name: ____________________	**Class/Batch:** ____________________
Father's Name: ____________________	**Gender:** Ⓜ Ⓕ Ⓞ
Roll Number: ____________________	**Date of Test:** ____________________

SECTION 1: SINGLE CORRECT CHOICE QUESTIONS (1 – 20)

1. Ⓐ Ⓑ Ⓒ Ⓓ	6. Ⓐ Ⓑ Ⓒ Ⓓ	11. Ⓐ Ⓑ Ⓒ Ⓓ	16. Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	7. Ⓐ Ⓑ Ⓒ Ⓓ	12. Ⓐ Ⓑ Ⓒ Ⓓ	17. Ⓐ Ⓑ Ⓒ Ⓓ
3. Ⓐ Ⓑ Ⓒ Ⓓ	8. Ⓐ Ⓑ Ⓒ Ⓓ	13. Ⓐ Ⓑ Ⓒ Ⓓ	18. Ⓐ Ⓑ Ⓒ Ⓓ
4. Ⓐ Ⓑ Ⓒ Ⓓ	9. Ⓐ Ⓑ Ⓒ Ⓓ	14. Ⓐ Ⓑ Ⓒ Ⓓ	19. Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	10. Ⓐ Ⓑ Ⓒ Ⓓ	15. Ⓐ Ⓑ Ⓒ Ⓓ	20. Ⓐ Ⓑ Ⓒ Ⓓ

SECTION 2: INTEGRAL/NUMERICAL TYPE QUESTIONS (21 – 25)

Write the final calculated single integer/numerical value inside the designated answer box clearly.

Q.21 Answer: ☐ **Q.22** Answer: ☐ **Q.23** Answer: ☐

Q.24 Answer: ☐ **Q.25** Answer: ☐

Signature of the Candidate **Signature of the Invigilator**